

STUDENT GRADE MANAGEMENT SYSTEM

PROJECT STATEMENT

1. Problem Statement

Managing student academic records manually can become difficult when there are many students, subjects, marks, assignments, and attendance records to handle. Keeping all this information organized on paper or in separate places can lead to mistakes and make it harder to find a student's performance quickly.

The Student Grade Management System is developed using Python to provide a simple way to manage these academic details. The system allows users to store student information, subjects, marks, grades, attendance, assignments, results, and overall performance in an organized manner.

2. Scope of the Project

The project covers the basic academic management requirements of a student. It includes student information, subjects, marks, grades, attendance, teacher information, examination results, assignments, and overall performance.

3. Target Users

- Students
- Teachers
- Beginners learning Python programming
- Schools and colleges for demonstration purposes
- Students developing academic programming projects

4. Objectives

1. Create a simple student academic management system.
2. Store and manage student information.
3. Record and calculate marks and grades.
4. Calculate attendance percentages.
5. Maintain assignment information.
6. Generate basic result information.
7. Calculate overall academic performance.
8. Demonstrate modular programming in Python.
9. Keep the project simple and beginner-friendly.

5. High-Level Features

Student Management: Add, view, update, delete, and search student information.

Subject Management: Store subject names, codes, credits, and teacher details.

Marks Management: Record internal, midterm, and final marks and calculate total marks.

Grade Management: Calculate grades according to percentage.

Attendance Management: Calculate attendance percentage from classes attended.

Teacher Management: Maintain basic teacher information.

Result Management: Calculate percentage and pass/fail status.

Assignment Management: Track assignments, marks, and submission status.

Performance Management: Calculate overall performance.

6. Technologies and Concepts Used

- Python 3
- Visual Studio Code
- Python lists
- Functions
- Conditional statements
- Loops
- User input and output
- Basic calculations
- Python modules
- Menu-driven programming

7. Data Management

The current system uses Python lists for storing information temporarily. It does not use MySQL, SQLite, text files, external databases, or cloud storage. Therefore, information entered during a program session is lost when the program is closed. This approach keeps the project simple for beginners.

8. Grading System

Percentage	Grade
90% and above	A+
80% – 89%	A
70% – 79%	B+
60% – 69%	B
50% – 59%	C
40% – 49%	D
Below 40%	F

9. Performance Calculation

The performance module considers average marks, attendance percentage, and assignment performance.

Overall Performance = (Average Marks × 60%) + (Attendance × 20%) + (Assignment Average × 20%)

10. Expected Outcome

After completing the project, the user should be able to add and manage student records, manage subjects, enter and calculate marks, calculate grades, record attendance, maintain assignments, manage

teacher information, view results, calculate overall performance, and understand how multiple Python modules work together.

11. Limitations

- Data is temporary and is lost when the program closes.
- There is no database connection.
- There is no login or authentication system.
- The system uses a command-line interface.
- The project is mainly designed for educational and demonstration purposes.

12. Future Scope

- MySQL or SQLite database integration
- Tkinter graphical user interface
- Login and authentication
- Student report card generation
- PDF result generation
- Performance charts and graphs
- Permanent data storage
- Backup and restore
- Search and filtering improvements
- Separate teacher and student accounts

13. Conclusion

The Student Grade Management System provides a simple way to manage important academic information using Python. The project demonstrates how a larger application can be divided into smaller modules, with each module handling a specific task. It is a beginner-friendly project for practicing functions, lists, loops, conditions, calculations, and modular programming, while also providing a foundation that can later be extended with a database and graphical interface.